

Warehouse: The End-to-End Pipeline Process

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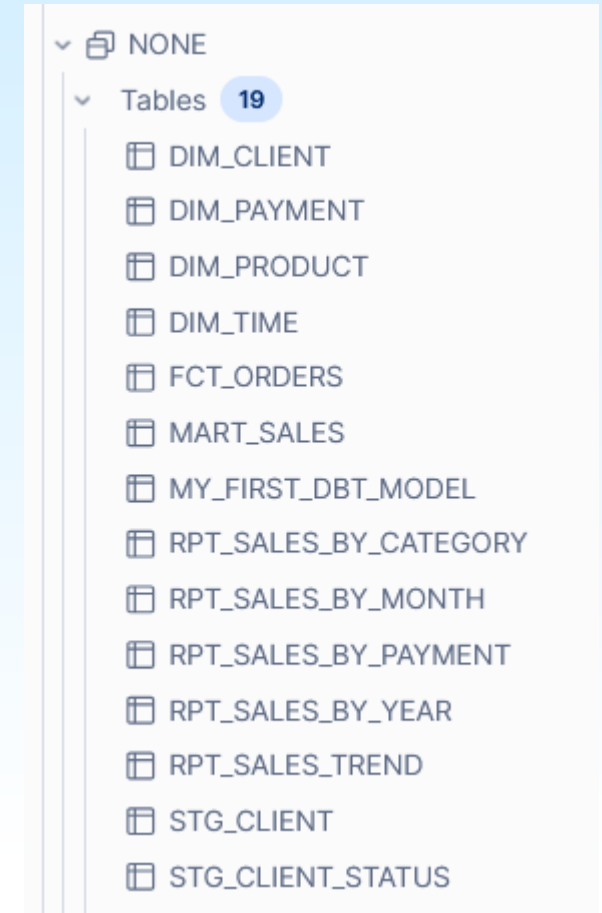
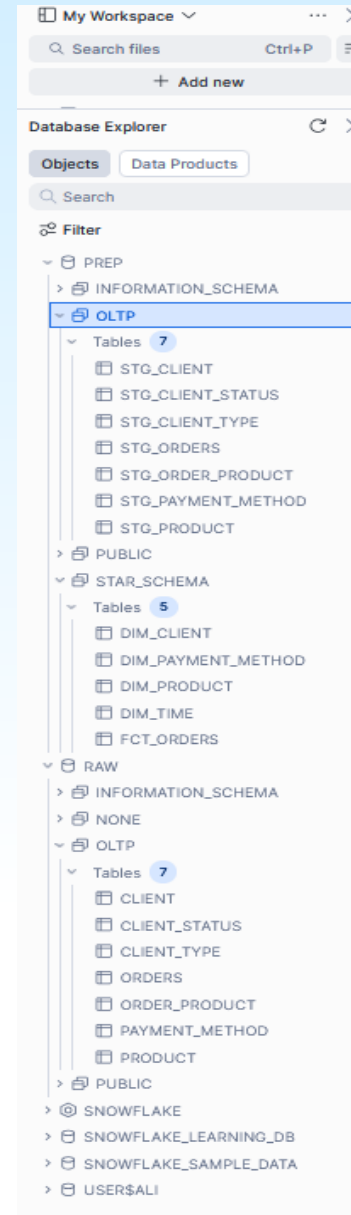
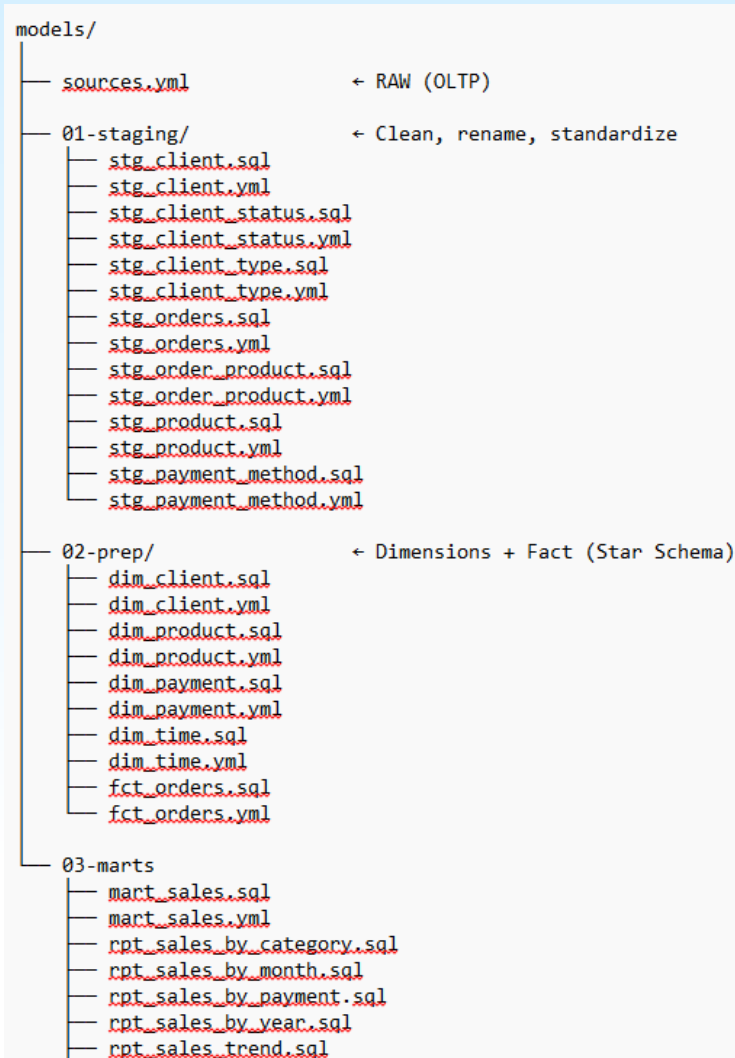
Project of Week 13

13.2.2026

Introduction

End-to-End Data Warehouse with Snowflake & dbt

Project Structure



End-to-End Data Workflow

Step 1: Source Data Analysis

- Identified operational tables (clients, orders, products, payments).
- Reviewed data types, keys, and date formats in Snowflake.

Step 2: Staging Layer (dbt)

- Created **staging models** to clean and standardize raw data.
- Renamed columns, fixed date formats, and removed inconsistencies.

Step 3: Dimensional Modeling

- Built **dimension tables** (Time, Client, Product, Payment).
- Created **fact table (FCT_ORDERS)** with sales metrics.
- Applied **star schema** design for analytics.

Step 4: Data Quality & Governance

- Added **dbt tests** (not-null, relationships).
- Anonymized sensitive client data using **dbt macros (hashing)**.

Step 5: Data Mart Creation

- Built **Sales Mart** optimized for reporting.
- Aggregated sales by time, product, and payment method.

Step 6: Reporting Tables

- Created reporting tables:

Step 7: Visualization & Insights

- Exported results to Python.
- Created **2D heatmap** and **3D sales plots**.
- Analyzed trends, seasonality, and business performance.

```
OLTP SYSTEM (Snowflake RAW)
|
├─ client
├─ orders
├─ order_product
├─ product
├─ payment_method
|
▼
DBT STAGING LAYER (models/01-staging)
|
├─ stg_client
├─ stg_orders
├─ stg_order_product
├─ stg_product
|
▼
DBT PREP / STAR SCHEMA (models/02-prep)
|
├─ dim_client      (anonymized)
├─ dim_product
├─ dim_payment_method
├─ dim_time        (generated by macro)
|
├─ fct_orders
|
▼
DATA MART (models/03-marts)
|
├─ mart_sales
|
▼
```

```
OLAP REPORTING TABLES
|
├─ RPT_SALES_BY_YEAR
├─ RPT_SALES_BY_MONTH
├─ RPT_SALES_BY_CATEGORY
├─ RPT_SALES_BY_PAYMENT
├─ RPT_SALES_TREND
|
▼
VISUALIZATION & INSIGHTS
|
├─ Line charts
├─ Bar charts
├─ Heatmap (2D OLAP)
├─ 3D Cube plot
```

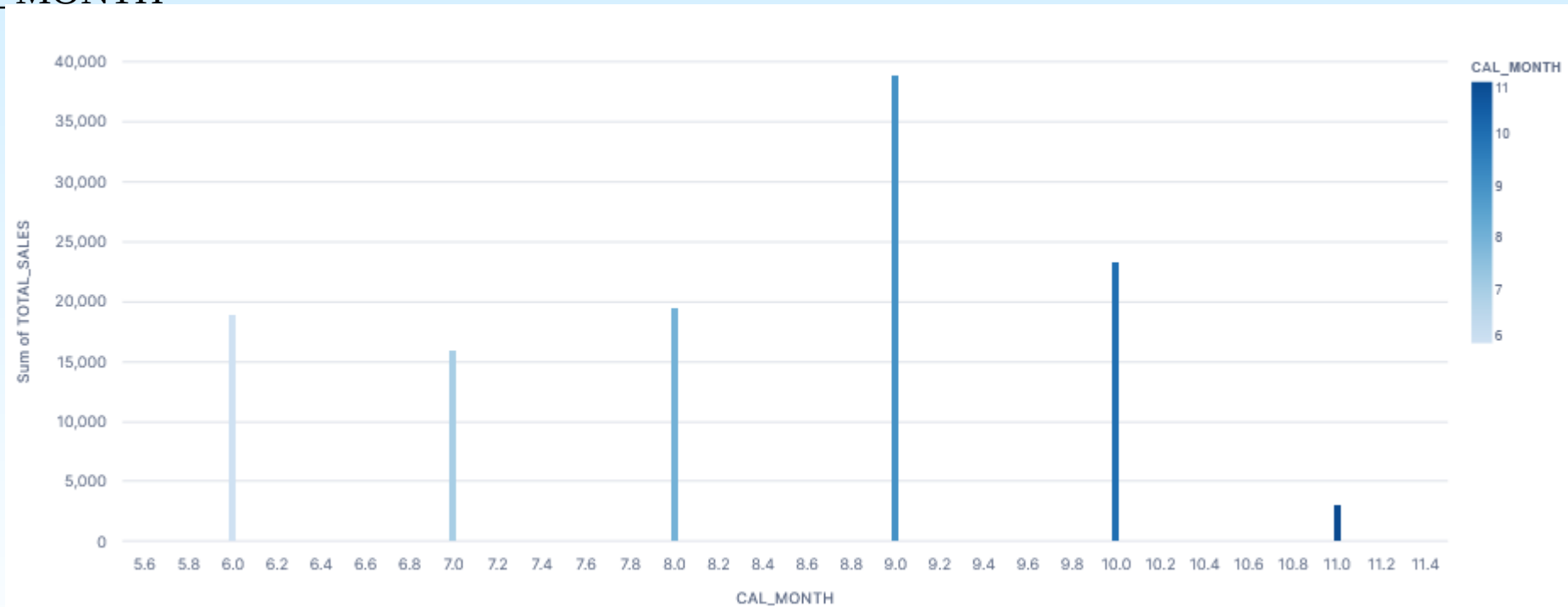
Total sales by month and year (OLAP)

Table name : RPT_SALES_BY_MONTH

Sql:

```
SELECT  cal_year,  
        cal_month,  
        total_sales  
FROM    RPT_SALES_BY_MONTH  
ORDER BY  
        cal_year,  
        cal_month;
```

This query presents total sales by month and year to highlight seasonal and temporal sales trends.



By analyzing total sales per month and year, this chart helps identify peak seasons and periods of growth or decline.

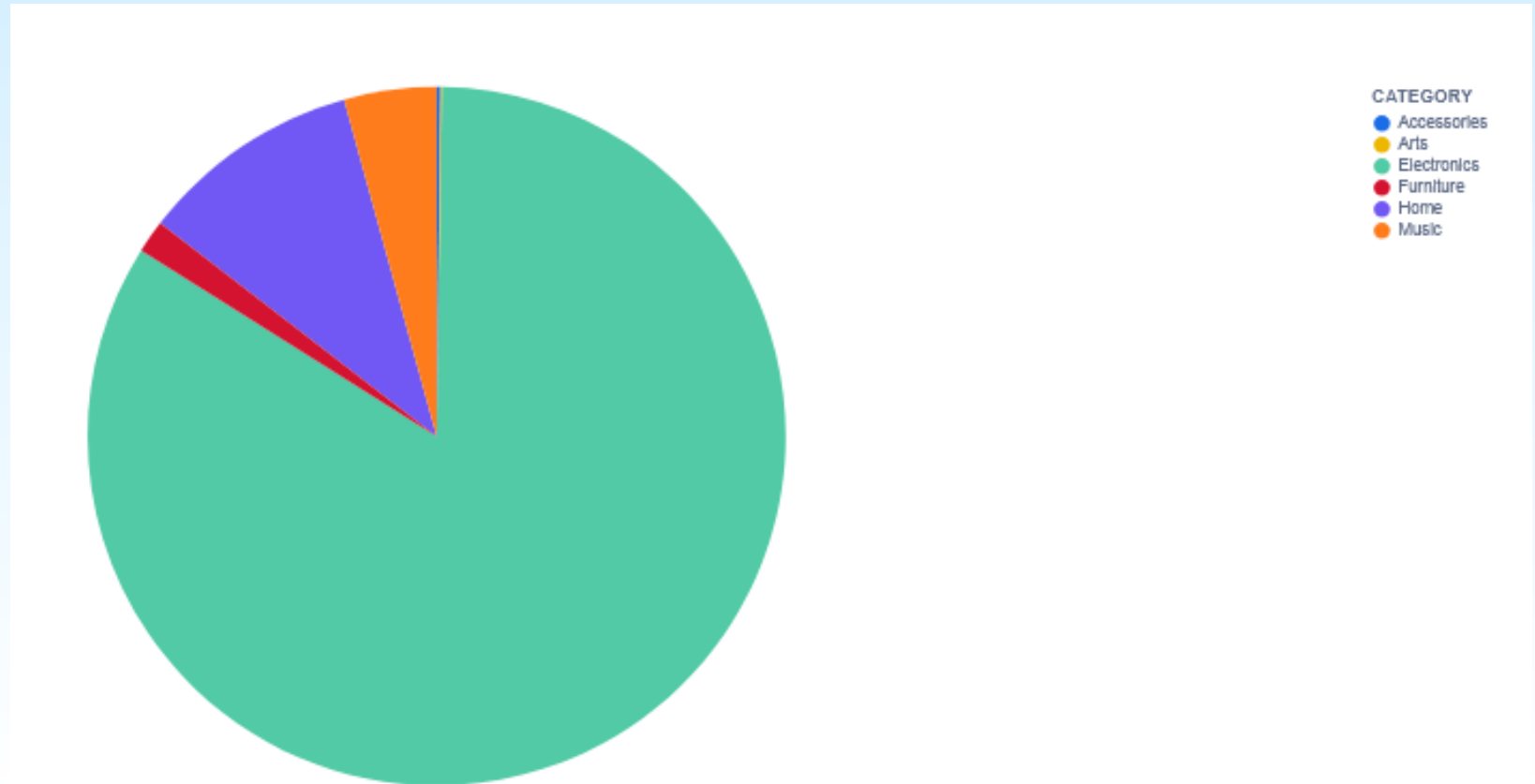
Total sales across product categories (OLAP)

Table name: RPT_SALES_BY_CATEGORY

Sql:

```
SELECT  category,  
total_sales  
FROM  
RPT_SALES_BY_CATE  
GORY  
ORDER BY  
total_sales DESC;
```

This query compares total sales across product categories, highlighting which categories contribute the most to overall revenue



By ranking product categories by total sales, this analysis identifies high-performing categories and opportunities for portfolio optimization.

Total sales by payment method (OLAP)

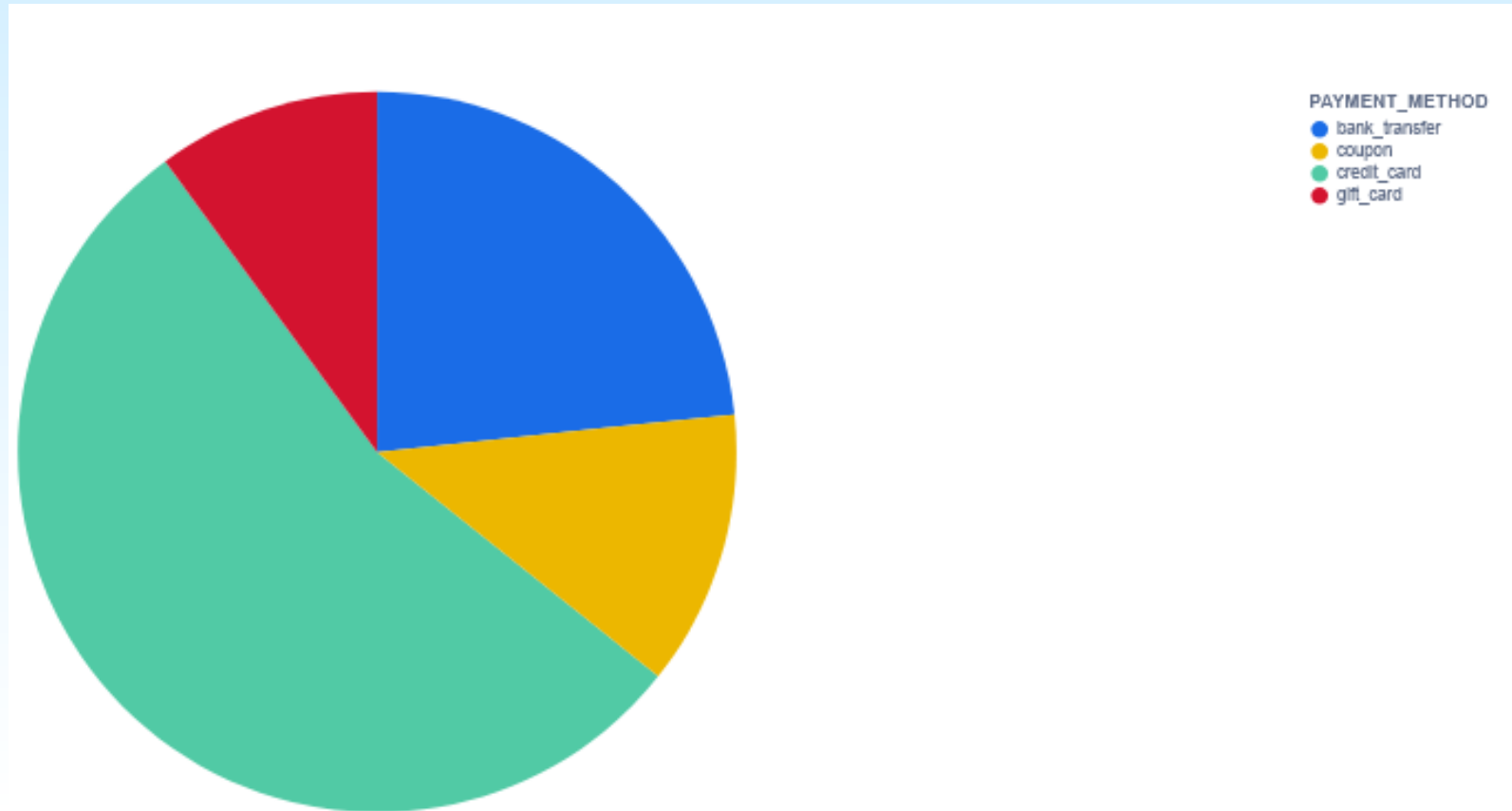
Table name

RPT_SALES_BY_PAYMENT

Sql:

```
SELECT payment_method,  
total_salesFROM  
RPT_SALES_BY_PAYMENTORDER BY  
total_sales DESC;
```

This analysis shows total sales by payment method, revealing which payment options customers prefer.

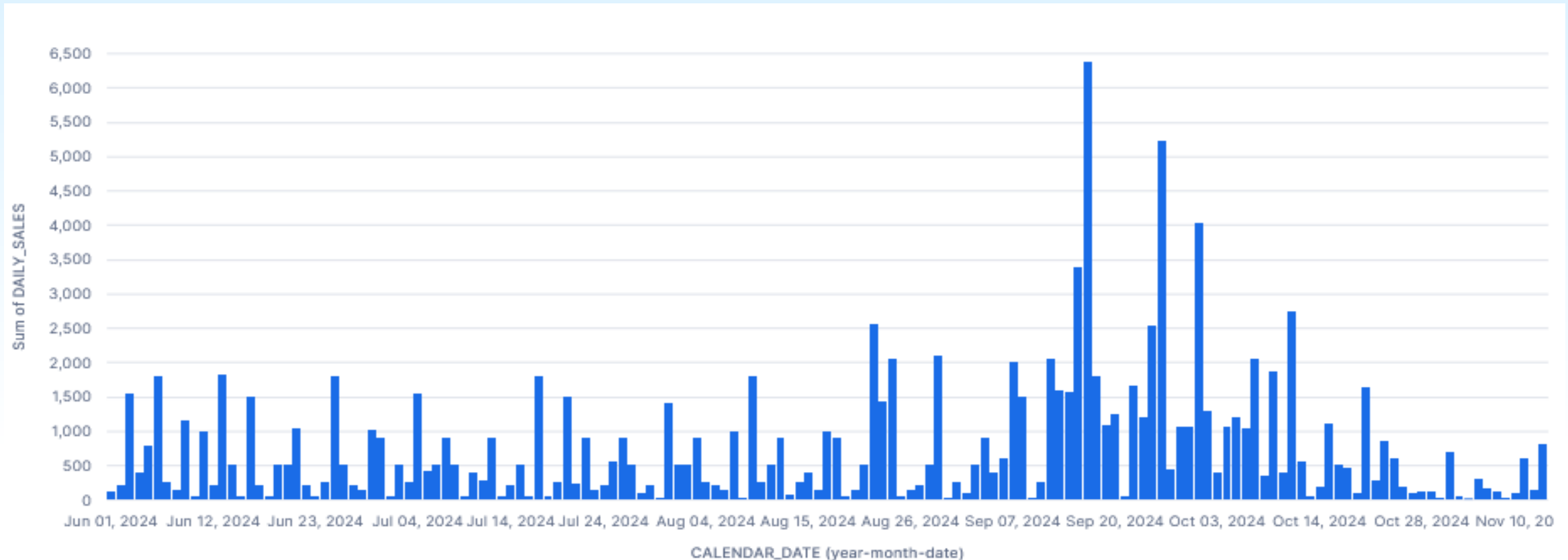


This chart compares total sales across different payment methods

Sales Trend Over Time (OLAP)

Table name: RPT_SALES_TREND

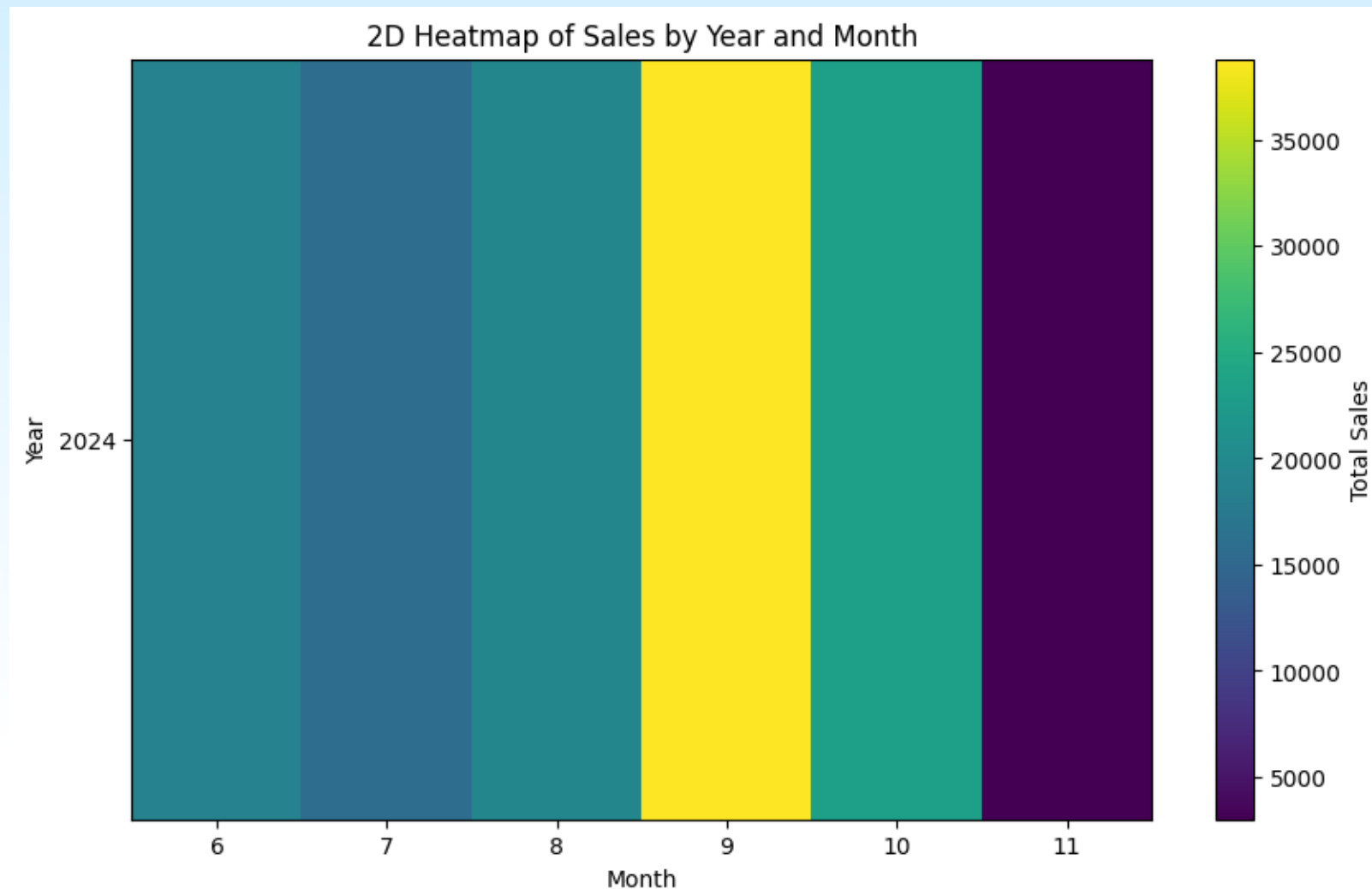
```
SELECT  calendar_date,  daily_sales
FROM    RPT_SALES_TREND
ORDER BY calendar_date;
```



This chart shows the daily sales trend over time, helping to identify growth patterns, seasonality, and demand fluctuations.

Monthly Sales Summary: 2D HEAT MAP (OLAP)

Sql:
SELECT cal_year, cal_month,
total_salesFROM
RPT_SALES_BY_MONTHORDER BY
cal_year, cal_month;



This visualization summarizes total sales by month and year, enabling comparison of monthly performance across different periods.

Summary

- This project transforms OLTP e-commerce data into an OLAP-ready sales data warehouse with governance, reporting, and analytical insights.
- Architecture: OLTP → Staging → Star Schema → Sales Mart → OLAP Reporting
- OLAP Reporting Tables:
 - Pre-aggregated tables for fast analytics:
 - RPT_SALES_BY_YEAR
 - RPT_SALES_BY_MONTH
 - RPT_SALES_BY_CATEGORY
 - RPT_SALES_BY_PAYMENT
 - RPT_SALES_TREND

Acknowledgements:

Moushumi and Rami for support

Thanks for your attention